

Provably Safe Policy Optimisation

Policies that stay safe when the shield comes off

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Shielded RL reports safety 1.00. Whose safety is it?

A **runtime shield** H overrides unsafe actions before they reach the environment. Training is safe and the deployed system scores a perfect safety rate.

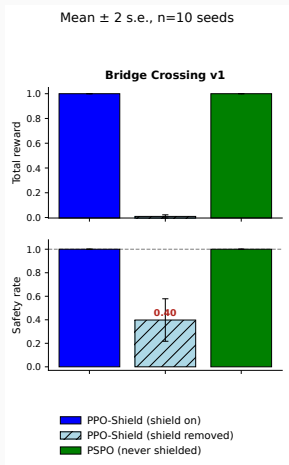
But the shield keeps rescuing the policy, so the policy never has to learn to be safe. Take the shield away and you find out what was actually learned.

Bridge Crossing v1, the same trained weights:

- shield on: reward 1.00, safety 1.00
- shield removed: reward 0.01, safety 0.40

The safety belonged to the wrapper, not to the weights.

The question: does the policy stay safe under *unshielded* greedy execution?



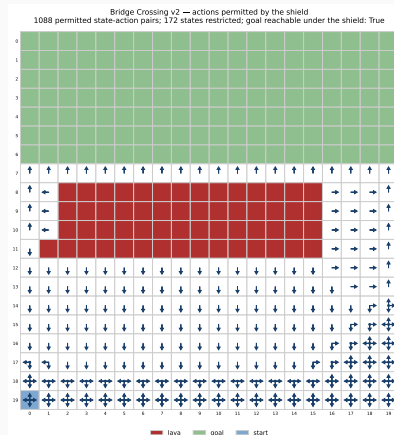
PSPO: two mechanisms, only one is needed at deployment

Shield H protects training-time *interaction*. Used only while collecting rollouts — never at evaluation.

LID Θ_k a certified region of *parameter* space that makes the exported greedy policy safe with no shield attached.

Step 0 — start from a safe policy. Turn the shield into a reward-free supervised dataset over every certified state and fit θ_0 with safe-set cross-entropy, an all-safe margin hinge $[m - d_s]_+$ ($m = 2$), and a safe-action entropy term. Accept θ_0 only if the greedy action is safe on every row, $\min_s d_s \geq 2$, and normalised safe entropy ≥ 0.95 ; otherwise abort.

The entropy term matters: before reward is observed there is no basis for committing to one safe action among several.



What the shield encodes: permitted actions per state.

Certify a region, not a point

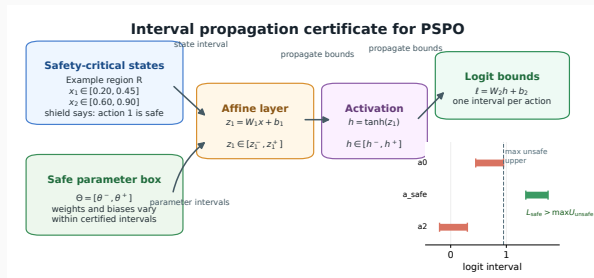
Interval bound propagation pushes a whole *parameter box* $\Theta = [\theta^-, \theta^+]$ through the network, giving per-action logit bounds valid for every $\theta \in \Theta$.

Θ is certified when, at every enumerated state,

$$\min_{a \in A_{\text{safe}}} \underline{z}_a(s; \Theta) > \max_{b \notin A_{\text{safe}}} \bar{z}_b(s; \Theta).$$

Then *every* parameter vector in the box picks a shield-admissible greedy action. Safety becomes a property you carry around, not a check you repeat each step.

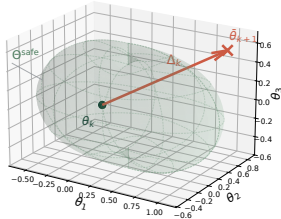
Stronger than “some safe argmax”: it separates *all* safe logits from all unsafe ones, leaving PPO room to move mass among safe actions.



The update: propose $\rightarrow$ grow $\rightarrow$ project

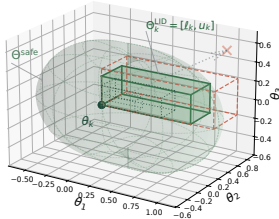
(a) Propose an update $\bar{\theta}_{k+1} = \theta_k + \Delta_k$

PPO proposes an unconstrained step Δ_k . It leaves the safe parameter set $\Theta^{\text{safe}} = \{\theta : H(s, \mu_\theta(s)) = 1 \forall s \in \mathcal{D}_{\text{sh}}\}$, which is curved and not axis-aligned.



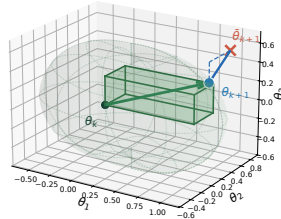
(b) Grow the maximal LID along Δ_k

Grow an axis-aligned box anchored at θ_k into the octant of Δ_k , until the IBP certificate fails (dashed). The LID is a certified inner approximation: $\Theta_k^{\text{LID}} \subseteq \Theta^{\text{safe}}$.



(c) Project $\bar{\theta}_{k+1}$ onto the LID

$\theta_{k+1} = \text{Proj}_{\Theta_k^{\text{LID}}}(\bar{\theta}_{k+1})$ clips coordinate-wise:
 θ_2, θ_3 hit a face, θ_1 keeps its proposed value.



- proposed update Δ_k
- safe set Θ^{safe} (curved, not axis-aligned)
- growth checkpoints
- coordinate-wise projection
- × uncertified proposal $\bar{\theta}_{k+1}$
- certified LID Θ_k^{LID} (axis-aligned box)
- over-grown box (fails certificate)
- accepted update $\theta_k \rightarrow \theta_{k+1}$

Each PPO iteration k : collect a *shielded* rollout; PPO proposes $\bar{\theta}_{k+1} = \theta_k + \Delta_k$; grow the maximal LID from θ_k along Δ_k until the certificate fails; then either project (coordinate-wise clipping) or **revert** to θ_k — fail closed.

Theorem 1

If θ_0 satisfies the invariant, and every enforcement either projects into a certified box or reverts, then every enforced iterate is greedy-safe on all certified states.

Experimental setup

Environment	States $\times$ Actions	Budget
Media Streaming	462×2	25k
Colour Bomb v1	81×5	25k
Colour Bomb v2	900×5	100k
Bridge Crossing v1	400×5	200k
Bridge Crossing v2	400×5	1.6M
MiniPacman [†]	9248×5	2.0M

Six methods: PPO, PPO-Lagrangian, PPO-PID-Lagrangian, CPO, PPO-Shield, **PSPO**.

[†] **PSPO** only so far — see the MiniPacman slide.

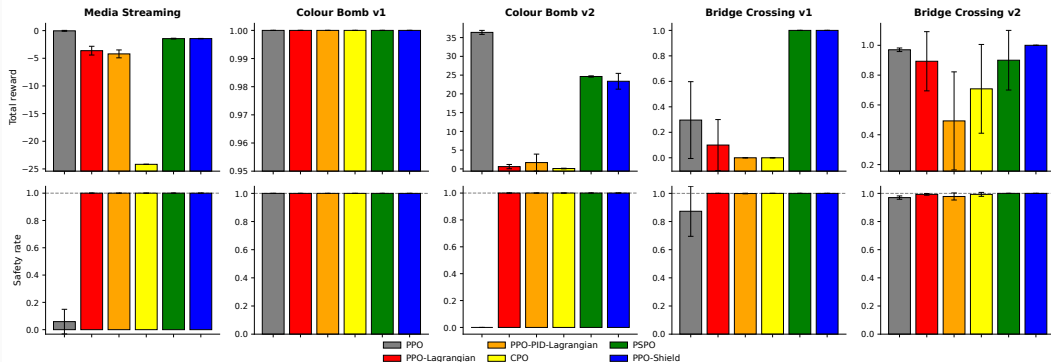
Protocol

- 10 seeds per method
- final greedy policy, 100 deterministic episodes per seed
- evaluated **unshielded** unless the method name says otherwise
- mean $\pm$ 2 standard errors throughout

Bridge Crossing v2 baselines and **PSPO** are both at 1.6M steps, so that comparison is budget-matched. On MiniPacman only **PSPO** has run at the extended budget so far.

Final greedy policy: reward and safety

Final greedy policy: total reward and safety rate (mean $\pm$ 2 s.e., n=10 seeds)

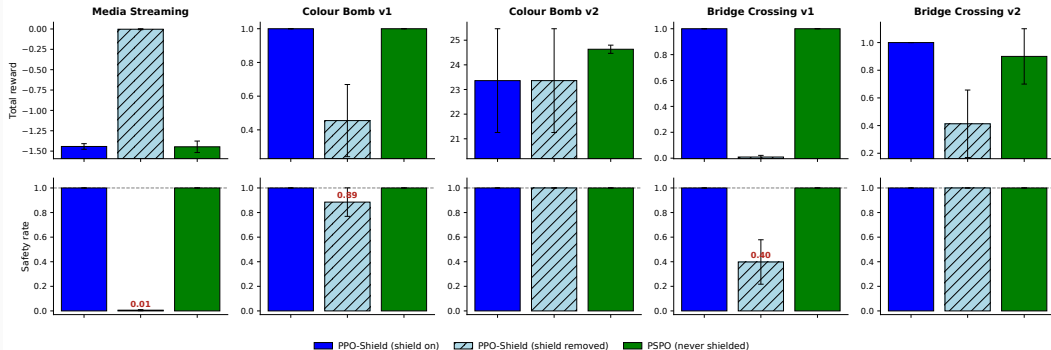


- Unconstrained **PPO** wins on reward exactly where it is unsafe: Colour Bomb v2 36.39 at safety 0.00; Media Streaming safety 0.06.
- **CPO** is safe but sits on the reward floor (−24.17 on Media Streaming) and is saturated.

- **PSPO** is the best *safe* method on Colour Bomb v2 (24.63 vs PPO-Shield's 23.36) and ties the best on Media Streaming (−1.45 vs −1.44).
- Reward bars are anchored at each panel's minimum, not zero.

Take the shield off

Whose safety is it? Same weights, evaluated with and without the shield (mean $\pm$ 2 s.e., n=10 seeds)

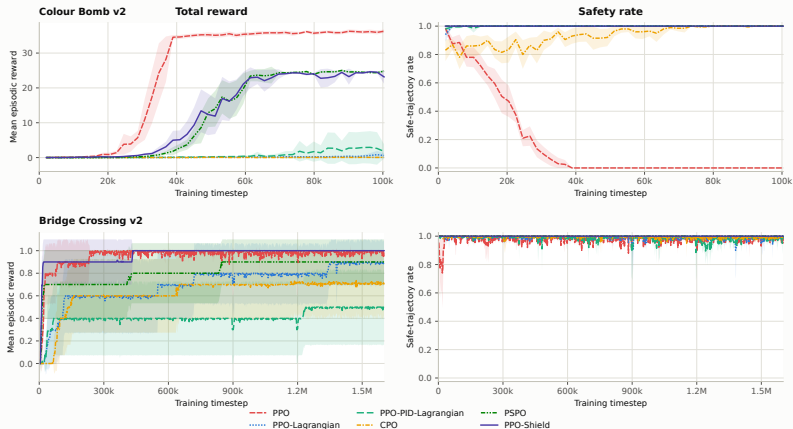


Same trained weights, with and without the runtime shield:
Media Streaming safety 1.00 $\rightarrow$ **0.01**; Bridge Crossing v1
safety 1.00 $\rightarrow$ **0.40** and reward 1.00 $\rightarrow$ **0.01**; Bridge Crossing
v2 reward 1.00 $\rightarrow$ **0.41**.

PSPO is never shielded at evaluation and matches the shielded numbers. Colour Bomb v2 is an honest null: the shield never binds there, so removing it changes nothing.

Evaluation-time learning curves

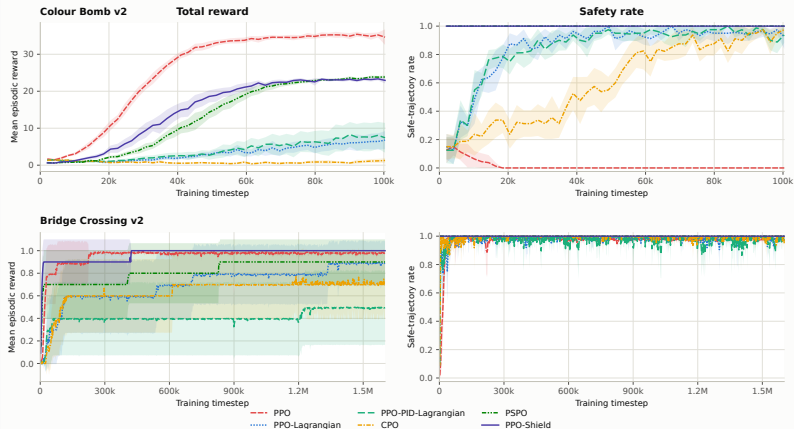
Evaluation-time learning curves — mean $\pm$ 2 SE across 10 seeds



Colour Bomb v2: PPO buys its reward by driving safety to zero by 40k steps, while **PSPO** and PPO-Shield climb with safety pinned at 1.00.

Exploration-time learning curves

Exploration-time learning curves — mean $\pm$ 2 SE across 10 seeds

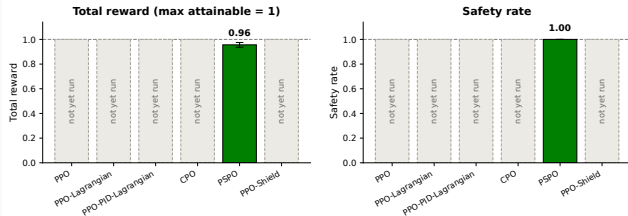


Safety during *training*, not only at convergence: the shielded rollout keeps every executed action admissible, so

PSPO pays no exploration-time safety cost to get there.

MiniPacman: PSPO nearly saturates the task

MiniPacman, 2M steps — PSPO only (mean $\pm$ 2 s.e., n=10 seeds); RL baselines outstanding



The largest domain here: 9248 states, 2M steps.

PSPO reaches reward 0.956 ± 0.019 at safety **1.000** — **95.6%** of the maximum attainable 1.00, unshielded.

Up from 0.839 ± 0.069 at the 500k budget: the earlier shortfall was a budget artefact, not a limit of the method.

Baselines outstanding. The extended-budget RL baseline sweep for MiniPacman has not produced results, so the placeholders are left explicitly empty rather than filled from a shorter run.

Where PSPO stands

What it buys

- Safety is a property of the *exported weights* — no shield rides along at deployment.
- Matches shielded PPO-Shield without the shield, where PPO-Shield's own policy falls to 0.40 safety.
- Bridge Crossing v2 solved seeds at a matched 1.6M budget: **PSPO 9/10** vs CPO 7, PPO-Lag 6, PID-Lag 4 — and unconstrained PPO **0/10**.
- MiniPacman: 0.956 of a maximum 1.00, at safety 1.000.

What it does not

- Does not win everywhere: Bridge Crossing v2 0.90 ± 0.20 vs PPO-Shield 1.00 ± 0.00 .
- The guarantee is deterministic and domain-specific: nothing is claimed outside the certified states, for a wrong shield, or for stochastic sampling.
- Projection can reduce or reject an update — the theorem is about safety, not reward.
- MiniPacman is not yet a comparison: its RL baselines have not run at the extended budget.

Use a shield to learn safely; use a certificate to stay safe.